

6.1 FY 24 Whitepaper Proposal

Improving Generalization of Neural Networks for Physics Domains

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1. Objective

Deep learning (DL) techniques perform well on computer vision and language tasks but there are gaps in the field's research that inhibit the successful use of DL methodology on physics problems. We propose foundational research to enable neural networks (NNs) to perform well on physics tasks, across a broad spectrum of physical environments and boundary conditions (i.e., to generalize well). We will develop new methods that attain high accuracy in the large problem spaces that are characteristic of physical domains, and demonstrate their utility on three Navy-relevant physics tasks.

2. Motivation and Background

Trained NNs have performed well on computer vision (CV) and language tasks (e.g., medical diagnosis, self-driving vehicles, and Navy applications) because they generalize well (i.e., the ability to make accurate predictions across a task domain, even for examples not observed during training).

In recent years, researchers have been incorporating neural networks (NNs) into physics emulators (e.g., Universal Differential Equations (UDE) [1], Physics-informed Neural Networks (PINN) [2, 3], and Neural Ordinary Differential Equations (NODE) [4]) and have demonstrated numerous benefits, such as orders of magnitude speedups, lower implementation costs, the ability to shape partial differential equations (PDEs) to better conform with observations, and several other benefits [5]. The merger of scientific computing, scientific simulation, and neural networks is described as the emerging field of "Simulation Intelligence" [6].

However, applying NNs in physics task domains is still a relatively new research focus and additional basic research is required because numerical solutions to differential equations are expected to work **universally** for all environments, realistic physical relationships, and boundary conditions but currently **neural networks do not generalize well enough** to do this (i.e., trained DNN models tend to generate incorrect predictions when encountering even small changes to the conditions of a physics environment). This is an important limitation. The literature is filled with papers comparing solutions from new combinations of NNs and differential equations to numerical simulators for a handful of examples in a few scientific fields. However, there is no evidence in the literature that NN solutions provide good test performance over the entire range of physically realizable inputs. Furthermore, while accurate generalization is crucial, there have been no prior studies on generalization in physics domains, which is a gap that this proposal addresses.

Fundamental machine learning research cannot be performed in a vacuum as it requires large datasets to train and evaluate newly developed methods. Unfortunately, large-scale physics domain datasets are scarce. Thus, we propose to develop and demonstrate the utility of novel generalization techniques on three physics task domains that are the focus of NRL projects in acoustics, meteorology, and material science, respectively. These three projects will uniquely provide us with the required datasets for studying our generalization methods and enable high transition potential.

3. Technical Approach

We propose to address two research hypotheses to guide our research:

1. **H1:** Some of the properties and methods that improve the generalization of mature deep learning methods on computer vision and natural language processing tasks, will also improve generalization on physics tasks. (Section 3.2)
2. **H2:** The physics equations, symmetry, and invariances in each domain can be used to improve the generalization in that domain. (Section 3.3)

To test these hypotheses, there are two major technical challenges that must be addressed:

1. **TC1 – Identify the generalization techniques used in these mature DL methods that work well across physics tasks.** While there exist theoretical and empirical studies on generalization [7 -13], none pertain to training NN models in physics. We propose to conduct a large-scale investigation of generalization in Navy DL physics applications using data that is available only at NRL (due to distribution constraints).
2. **TC2 - Derive generalization methods that work well for our three physics domain testbeds.** Using the physics equations between physical and derived quantities, we plan to develop and evaluate techniques to improve network generalization in the large problem space of Navy physics domains – both processes that work across domains and domain-specific methods.

In recent years, interest in integrating NNs with physics simulators has grown exponentially but the technical challenges and hypotheses listed above have not yet been addressed. In addition, the research proposed here can be performed only at NRL due to the classification of the data and the relevance of these domains to the Navy.

Throughout this project, we will develop novel architectures and methods, evaluate them, and report on our results. Although we are proposing 6.1 research, the success of our work has high potential to transition and improve existing Navy applications used by the Fleet.

3.1 Task 1: Create baselines

We will build on current efforts to implement NN surrogates for three task domains of interest to NRL researchers: acoustic modeling, RF propagation, and structural health monitoring.

Acoustics: The performers of the 6.2 PALIS project (Smith and Fabre) are developing two neural network based methods for estimating transmission loss (TL) from ocean properties for limited ranges of ocean conditions and source depths/frequencies. Their implementations will be the baseline for the acoustics domain. Here, we propose to develop methods for training NNs such that they produce accurate 2D TL maps across time, and for a broad spectrum of ocean properties (i.e., temperature, salinity, sound speed), sound source depths, receiver locations, and sound source frequencies.

Meteorology: Estimating radio frequency (RF) propagation in the atmosphere has many similarities to estimating TL in the underwater acoustics domain, so we anticipate that the NN solutions developed in PALIS can be used with the Navy's Couple Ocean/Atmosphere Mesoscale Prediction System (COAMPS®) NWP data. The acoustics software will be modified for this data and tested in the new domain. This modified implementation will constitute our meteorological baseline. Then we will investigate modifying these algorithms to produce accurate RF propagation maps across location, time, across a spectrum of atmospheric properties (i.e., humidity, temperature, pressure), trapping conditions (sub-refractions, standard-refractions, super-refraction, and trapping), and relevant RF source heights, receiver locations, and RF source frequencies.

Structural health monitoring: This task domain differs substantially from the prior two domains. The performers for the 6.1 MINC project (i.e., Smith and Rudolf) are developing a real-time NN system for monitoring material damage on naval platforms. They are developing a system in a laboratory setting that we will use as the baseline for this domain. We propose developing NN training methods that enable the developed systems to perform with high accuracy over a large variety of metallic surfaces (e.g., steel, aluminum, composites), across all configurations of sensor locations, for complex structural geometries, interacting boundaries, non-linear wave dynamics (dispersion, interference, attenuation), and a near infinite population of possible damage characteristics.

Existing data and software will be collected and tested for the acoustic modeling, RF propagation, and structural health monitoring domains.

3.2 Task 2: Investigate neural network generalization in physics domains

We will investigate the importance of the properties that impact generalization for mature NN applications (such as computer vision and natural language processing tasks) that can be used in the context of our three physics domains – acoustic modeling, RF propagation, and structural health monitoring. Specifically, these properties are data, data augmentation, data diversity, architecture, non-linear activation functions, and hyper-parameters (H1).

Some of our early tasks in studying generalization will include defining measures, studying the DL literature on generalization, and understanding how physics impacts model design. Existing measures for studying generalization in CV fall within three categories: complexity, norm, and sharpness measures. However, these three types of measures are not equivalent and there

does not exist a comparison of these measures on physics ML models. Therefore, we propose to evaluate existing generalization measures in our three domains as part of our investigation. Transfer learning and unsupervised network pretraining are additional mechanisms used to improve generalization. Hence, we will investigate the use of these mechanisms for improving generalization in our domains (H1). That is, if there is a source domain with large amounts of data or other properties that enable good generalization ability, we will assess whether starting training with those pretrained networks retains some or most of that generalization ability after fine-tuning in the target domain.

Appropriately applying NNs to physics tasks requires a good understanding of deep learning techniques, the data, the domain's physics, and relevant differential equations. The form of these equations can guide the design of the architecture. In addition, the physics equations can be used to determine what data augmentation to use during training. Hence, part of our efforts will be on gaining additional expertise in each domain.

3.3 Task 3: Improving generalization for physics domains

We will leverage the insights obtained from investigating generalization for three Navy task domains and to invent novel techniques for improving generalization based on the physics equations, symmetries, and invariances in these domains (H2).

Many data augmentation methods used for mature deep learning tasks are inappropriate for physics modeling (i.e., one can't use data augmentation techniques that change physical properties without impacting the desired solution). Thus, we plan to investigate promising data augmentation methods for our three domains. For example, two existing data augmentation methods (i.e., cutout [14] and patch masking [15]) hide parts of the input but requires the network to learn to produce the entire output field. Although these methods have not been used in physics NN networks, their success in CV leads us to believe they will improve generalization across physics domains too.

In addition, we plan to invent novel ways of performing data augmentation within the constraints of the domain's physics. This will require understanding the symmetries and invariances to develop automatic techniques for augmenting the data that can be incorporated into the model by way of the network architecture and non-linear activation functions, which impacts the generalization ability of trained models (H2). We plan to utilize methods from Neural Architecture Search (NAS) and Hyper-Parameter Optimization (HPO) to allow an automated search of many combinations of potential architectures and activation functions. This task will directly benefit from the experiments developed in Task 1 since the measures and experiments will be employed. Furthermore, the PI's previous work in cyclical training of neural networks [16, 17] can be used to invent new dynamic activation functions that optimally adapt during model training.

Next, we provide an example of a novel loss function as an emblematic method for improving generalization. A common goal for replacing computationally expensive components in physics

simulators is to emulate the output of a physics simulator but orders of magnitude faster. In this case, it is common to use a mean-squared error (MSE) loss function to train the network to match the component's output. Here we propose an alternate loss function that supplements the MSE loss term with negative loss terms (i.e., for the network to learn not only what the output should be but also what it is not).

This type of loss has similarities to at least three well-known loss functions. The first is cross-entropy softmax, which is standard for classification problems. This loss function has both positive terms (i.e., which says "this is the correct label") and negative terms (i.e., which says "do not predict the wrong labels"). The second is unsupervised contrastive learning, which contains both positive terms (e.g., features of two augmentations of the same image should be the same) and negative terms (e.g., features of two images should be different). The third comes from metric learning, such as lifted structure feature embedding [18]. Here too the network is trained with both positive and negative loss terms.

We name our novel loss "contrastive output augmentation" as we augment the positive MSE term (i.e., which encodes that "every pixel in the output from the network should be the same as every pixel for the ground truth output") with negative terms (i.e., which encodes that "every pixel in the output from the network should differ from every pixel of other outputs") in a way that resembles cross-entropy softmax. Specifically, the positive MSE loss is defined as:

$$L_i = -\log [|| O_i - f(x_i) ||^2],$$

where O_i is the output from the simulator for the i^{th} input, x_i , and $f(x_i)$ is the equivalent output from the network. Our loss also defines

$$L_i = + \sum_j \log [|| O_j - f(x_i) ||^2]$$

where the j^{th} outputs, O_j , can be the simulator outputs for different inputs. This introduces negative examples during training that add greater variety to each training epoch that will improve generalization. Developing methodologies like this and others will be one of the goals of Task 3.

3.4 Evaluation

As described in Section 3.1, we will develop baselines for all three domains. Task 2 includes defining generalization measures to evaluate existing generalization methods in physics domains. The majority of our effort for the generalization study will involve defining/executing experiments and analyzing the results (H1). The purpose of these experiments will be to accurately measure the generalization ability of DNNs along the dimensions of importance to our application domains. For example, a potentially worthwhile experiment would be to test the sensitivity of our acoustics algorithms to small changes in source frequency and location.

Promising methods and techniques developed in Tasks 2 and 3 will be evaluated by our team's three subject matter experts (SME). Deliverables will include methods that improve generalization across many physics domains (H1), a methodology for designing methods to improve generalization based on the domain's physics (H2), and updated implementations in our three domains. Since our goal is to improve the performance of algorithms that are used

on Navy platforms, we expect that our team's SMEs will be able to transition improvements to these algorithms, as appropriate.

4. Plans

FY24: We will create our baselines (Task 1) and conduct a large-scale investigation of methods for improving generalization from mature DL applications in our three physics domains, with the implementations and data available at NRL (Task 2). We will assess whether H1 holds.

FY25: We will leverage the insights obtained from investigating generalization for acoustic modeling, RF propagation, and structural health monitoring to enable NN versions of these physics simulators to make accurate predictions across a wide range of environments and conditions (Task 3). We will design network models based on the physics domain equations and symmetries, and then evaluate them with respect to generalization within their domain (H2/Task 3).

FY26: We will continue our experiments with generalization for acoustic modeling, RF propagation, and structural health monitoring by inventing and testing novel techniques in these domains. Our team will transition incremental improvements to existing methods to PALIS, COAMPS-OS, or Builder software, as appropriate.

Throughout this project we will publish papers that describe our progress and results.

5. References

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6. Funding Profile

Expense	FY24 (\$K)	FY25 (\$K)	FY26 (\$K)
PI: Leslie N. Smith, 5514	236	248	260
Josie Fabre, 7180	80	62	65
Andrew J. Kammerer, 7542	50	55	60
Christopher Rudolf, 6354	50	55	65
2 NREIP summer interns	20	22	24
Travel and equipment	5	6	7
Total	441	448	481

7. Budget Justification

The budget pays for 50% of the PI, Leslie N. Smith (5514), 25% of the time of Josie Fabre, who is an SME in acoustic modeling, 15% of the time of Andrew J. Kammerer, who is an SME in RF modeling in the atmosphere, and 15% of the time of Christopher Rudolf, who is an SME in structural health monitoring.

8. Participant Information

This project requires active participation by four NRL Divisions: Information Technology, Acoustics, Meteorology, and Material Sciences. Subject matter experts (SME) in each of the physics domains are necessary for generating the data, specifying the requirements of the physics applications, and evaluating predictions produced by new training methods. SMEs in

deep learning are necessary for investigating neural network generalization in these domains, designing/implementing new methods, and executing robust sets of experiments.

Dr. Leslie N. Smith (PI at 50%) received a combined BS and MS degrees in Physical Chemistry from the University of Connecticut in 1976 and a Ph.D. degree in chemical physics from the University of Illinois in 1979. He is currently at the US Naval Research Laboratory in the Navy Center for Applied Research in Artificial Intelligence. His research focuses on deep learning for computer vision and physics-based domains, utilizing his experience with convolutional neural networks, reinforcement learning, machine learning, semi-supervised learning, and compressed sensing. In addition, he hosts widely attended seminars on AI. His collaborations with Dr. Fabre on the PALIS Base program and Dr. Rudolf on the MINC Base program sets the groundwork for this proposed research.

Dr J. Paquin Fabre has 30+ years of experience in underwater acoustics and related disciplines and has ~62 publications. She was recently selected to receive the NDIA Undersea Warfare Bronze Medal award for her contributions to computational techniques for ocean acoustics which have revolutionized the process by which the Navy determines ocean propagation. She is the Acoustics division representative to ONRs Task Force Ocean. Her research focusses on improving accuracy and speed of environmental acoustic simulations for Naval applications. She heads and participates in several basic and applied research projects in areas of acoustic provincing, applications of machine learning for acoustics, data assimilation, environmental acoustic sensitivity, reconstruction and analysis, and acoustic simulation and sensor performance prediction. She has an excellent transition record for a number of ocean acoustic capabilities.

Dr. Chris Rudolf is a 2018 Karles Fellow and a recognized scientist focusing on developing new experimental methods for the unique analysis and characterization of multifunctional materials. Dr. Rudolf received his Ph.D. degree in 2016 from Florida International University where he developed novel in-situ experimental methods for characterization of material deformation and failure modes of graphene reinforced ceramics. He previously led the research on an NRL 6.1 program looking at electrically-assisted deformation of metals, where his experience and background in experimenting with state-of-the-art sensing systems and mechanical test systems was instrumental in moving the program forward. Currently, he is the PI on an NRL 6.1 program where he is leading the research into the use of machine learning for material-systems intelligence.

Mr. Andrew Kammerer (code NRL 7542) has expertise in physical oceanography and meteorology, with a B.S. and M.S. from Coastal Carolina University in 2014 and 2017 respectively. He has been employed with the NRL Marine Meteorology division since 2017. In that time his research focus has been on the impact of the atmosphere on EM RF propagation, forcing propagation modeling using NWP atmosphere, validation and improvements to Navy NWP and NWP forced propagation modeling, and efforts supporting operational EM propagation modeling including NWP data compression, cloud computing based propagation modeling, and improvements to EM propagation TDAs. He is PI on three ONR sponsored

projects (Codes 31 and 32) including two 6.3 and one 6.4 effort. He is also participating in several cross divisional 6.1 and 6.2 efforts sponsored by the NRL Base Program, NISE, and ONR.

9. Team Expertise

The members of Code 5514 have a wide variety of experience with basic and applied machine learning, with a special focus on creating ML surrogate models by embedding neural networks within physics models. They are expert programmers with backgrounds in science that enable our work in the hybrid domain combining material science and machine learning.

Other members of our team are subject matter experts in the three physics domains described in this proposal. As described above, they have years of experience with their domain's tasks, including using machine learning to improve them.